Q $y = \csc x \qquad y' ?$

A $y = \frac{1}{\sin x}$

$$y' = \frac{(\sin x)(0) - (1)(\cos x)}{\sin^2 x} = \frac{-\cos x}{\sin x \cdot \sin x} = -\frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} = -\cot x \csc x$$